Azure CNI network plugin

- Also known as advanced network plugin
- --network-plugin azure
- IP address assignment:
  - Nodes: from AKS subnet
  - Pods:
    - from AKS subnet (traditional)
      - IPs get preallocated based on each node based on maxPods parameter
        - 1 IP for the node + pod IPs (as per maxPods parameter)
    - from a separate pod dedicated subnet in AKS's VNET (dynamic allocation of IPs and enhanced subnet)
      - IPs are assigned from the pod dedicated subnet
  - Pod-to-pod communication: direct, with no additional hop
  - Pod to device inside the VNET: the device will see pod's IP as the source address
  - Pod to device outside the VNET: the device will see node's IP as the source address, but that can be changed by modifying the masquerade rules

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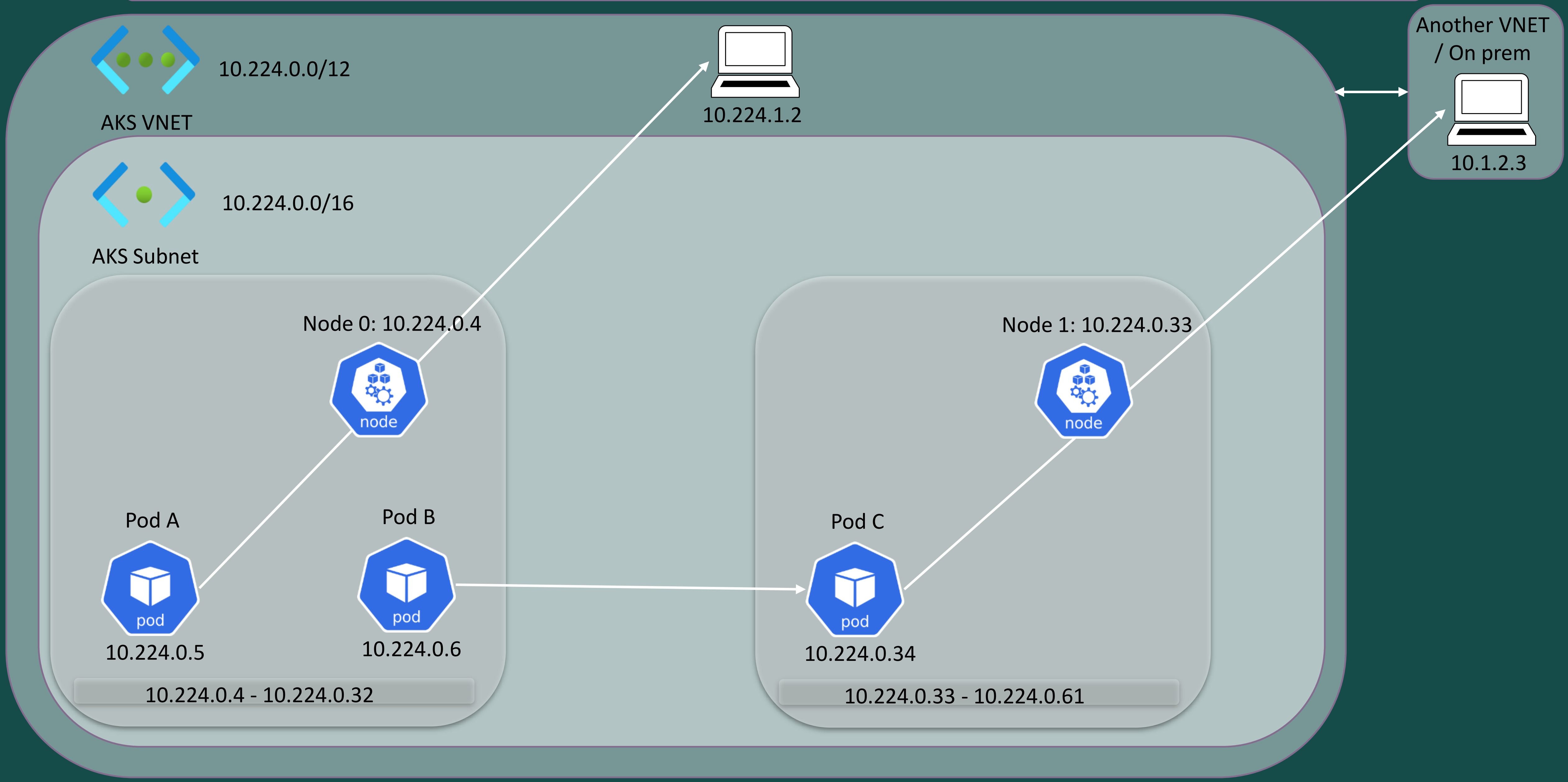

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- **Why would you choose Azure CNI?**
  - Pods receive IPs from the subnet -> Less hops -> Less latency
  - No need for managing UDRs
  - Communication happens mostly with devices outside of the cluster
  - Multiple clusters in the same subnet are allowed
  - You need features like Windows nodes, Azure Network Policy, Virtual Nodes

- **What disadvantages apply to Azure CNI?**
  - Requires many IP addresses and IP address planning
    - Can lead to IP addresses exhaustion if the cluster grows
    - Upgrade or other operations require at least one buffer node that needs IPs so you will always need maxPods+1 IPs for this operations

Mostly addressed by dynamic allocation of IPs and enhanced subnet which provides:

- Improved IP usage
- Flexibility and scalability
- High performance
- Others